



A
Project Synopsis
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Abstract

EcoBazaar is a sustainability-focused e-commerce platform designed to promote environmentally responsible consumption by providing users access to verified eco-friendly, biodegradable, and locally produced goods. The project addresses the growing global concern regarding the harmful impact of non-biodegradable materials, excessive plastic usage, and misleading “greenwashed” marketing practices that confuse environmentally conscious consumers. To overcome these challenges, EcoBazaar proposes a transparent digital marketplace where every listed product undergoes a structured eco-verification process based on material composition, biodegradability, carbon footprint, and sourcing practices.

The platform integrates modern web technologies with sustainability analytics to create a seamless and trustworthy user experience. It incorporates modules for eco-product categorisation, seller verification, sustainability scoring, and personalised green recommendations. These mechanisms ensure that users not only purchase environmentally safe products but also understand their broader ecological impact. Furthermore, EcoBazaar empowers small-scale artisans and cottage-industry manufacturers by providing them a low-cost, high-visibility digital space, thereby supporting inclusive and sustainable economic growth.

From a technical standpoint, the system adopts a scalable MERN-based architecture (MongoDB, Express.js, React.js, Node.js), secure authentication, and API-driven modularity. The integration of a sustainability dashboard enables users to track metrics such as estimated waste reduction and carbon offset, encouraging longterm behavioural change. By bridging the gap between eco-conscious consumers and authentic green manufacturers, EcoBazaar aims to evolve into a comprehensive ecosystem that promotes responsible consumption, environmental awareness, and sustainable lifestyle adoption.

Introduction

The rapid rise in environmental degradation, driven by the excessive use of plastics, non-biodegradable materials, and unsustainable manufacturing practices, has heightened global concern regarding the long-term impact of consumer behaviour. As individuals become more aware of climate change and ecological imbalance, the demand for eco-friendly and ethically produced goods has increased significantly. However, the absence of a unified and transparent digital marketplace for genuine eco-products creates confusion, reduces accessibility, and often leads to misleading sustainability claims. EcoBazaar is conceptualised as a solution to this gap—a dedicated e-commerce platform that exclusively promotes verified eco-friendly, biodegradable, and sustainable products.

Despite the increasing popularity of online shopping, mainstream e-commerce platforms do not prioritise sustainability as their core objective. Eco-friendly items, when offered, are often mixed with conventional products, making it difficult for users to identify and trust environmentally responsible options. Moreover, the issue of “greenwashing,” where sellers falsely label their products as eco-friendly without scientific validation, further complicates the decision-making process for conscious consumers. EcoBazaar aims to counter these limitations by implementing a structured eco-verification workflow, ensuring that every listed product meets predefined environmental standards. This approach not only builds user trust but also encourages informed and responsible purchases.

Another significant motivation behind EcoBazaar is the need to support small-scale artisans, rural producers, and cottage-industry manufacturers who create natural and sustainable products but struggle to reach larger markets due to technological and financial limitations. By providing an accessible digital platform with simplified onboarding and low operational barriers, EcoBazaar empowers these local businesses, enabling them to scale sustainably while preserving traditional craftsmanship. This aligns the platform with both environmental and socio-economic development goals, contributing to community upliftment. From a technological perspective, EcoBazaar integrates modern web development frameworks to deliver a reliable, scalable, and user-friendly shopping experience. The platform incorporates features such as personalized eco-friendly recommendations, sustainability scoring based on product materials and manufacturing processes, and dashboards that display environmental impact metrics such as carbon footprint reduction. These features position EcoBazaar not merely as a marketplace but as an educational ecosystem that fosters long-term behavioral change towards sustainability.

In essence, EcoBazaar combines environmental responsibility, technological innovation, and community empowerment into a single cohesive platform. By bridging the gap between eco-conscious consumers and authentic sustainable product sellers, it aspires to redefine digital commerce with transparency at its core. The project not only promotes sustainable consumption but also contributes to the broader mission of reducing environmental harm and supporting local economies. Through its unique blend of verification mechanisms, awareness tools, and modern e-commerce functionality, EcoBazaar represents a progressive step towards a more sustainable digital marketplace.

Rationale

The rationale behind developing EcoBazaar emerges from the growing environmental crisis caused by unsustainable consumption patterns and the widespread use of non-biodegradable materials. Despite increasing public awareness about environmental protection, consumers often lack easy access to genuinely eco-friendly products. Traditional marketplaces are saturated with mass-produced items that contribute to pollution and waste generation. This gap between environmental consciousness and product availability underscores the urgent need for a specialised platform that simplifies access to sustainable alternatives and encourages responsible purchasing habits.

The modern e-commerce landscape presents another challenge—a significant rise in “greenwashing,” where businesses falsely market products as environmentally friendly without scientific proof or regulatory backing.

Such misleading practices not only misinform consumers but also dilute the credibility of genuine eco-friendly manufacturers. EcoBazaar addresses this issue through a structured verification system, ensuring that products listed on the platform meet clear sustainability criteria. By enhancing transparency, the platform builds user trust and sets a higher standard for environmental responsibility within digital marketplaces.

An equally important rationale is the support and empowerment of small-scale artisans, rural producers, and cottage industries whose livelihoods depend on sustainable, handmade, or natural products. These producers often lack the marketing reach, digital infrastructure, and financial resources required to compete with larger brands on mainstream e-commerce platforms. EcoBazaar offers them an inclusive and accessible digital marketplace that prioritises authenticity over scale. This not only helps preserve traditional craftsmanship and local economies but also promotes decentralised, low-impact production models essential for long-term ecological balance.

Technologically, the project aligns with India's rapid digital transformation and increasing adoption of online shopping across urban and semi-urban regions. By integrating modern web technologies, sustainability scoring mechanisms, and eco-awareness tools, EcoBazaar uses digital innovation to solve real-world environmental challenges. The platform's design ensures that users are not only able to purchase eco-friendly products but are also educated about their environmental impact through intuitive dashboards and metrics such as carbon footprint reduction and material biodegradability.

Overall, the rationale for EcoBazaar is rooted in a combination of environmental necessity, consumer empowerment, technological innovation, and socio-economic inclusivity. It seeks to bridge the gap between conscious consumers and authentic sustainable product sellers while creating an ecosystem that promotes longterm ecological well-being. By addressing critical gaps in trust, accessibility, and awareness, EcoBazaar positions itself as a meaningful step toward a more sustainable and responsible digital economy.

Objectives

The primary goal of this project is to leverage modern web technologies and sustainability-focused analytics to create a transparent, user-friendly, and environmentally responsible e-commerce ecosystem. By integrating structured eco-verification, intelligent recommendation systems, and engaging awareness tools, EcoBazaar aims to empower consumers to make informed decisions while supporting authentic eco-friendly manufacturers. The following objectives outline the key targets for developing a reliable, scalable, and sustainability-driven digital marketplace:

Develop a dedicated eco-friendly e-commerce platform

that enables seamless browsing, comparison, and purchase of verified sustainable products, ensuring users have access to authentic biodegradable and environmentally safe alternatives across multiple categories.

Integrate an eco-verification and sustainability scoring system

to evaluate each product based on factors such as material composition, biodegradability, carbon footprint, and ethical sourcing, thereby providing transparent metrics that guide responsible consumer choices.

Implement personalised recommendation algorithms and impact analytics

to deliver user-specific sustainable product suggestions and visual dashboards that display environmental benefits, such as estimated waste reduction and carbon offset, enhancing user engagement and awareness.

Design and deploy a scalable, user-friendly web application

built with React.js/Next.js and a secure backend architecture, enabling smooth user experience, efficient seller onboarding, real-time inventory management, and secure payment processing across diverse digital environments.

These objectives are designed to ensure the development of a comprehensive, reliable, and sustainability-focused marketplace. Achieving these goals will advance the accessibility of eco-friendly products, strengthen trust through verification and transparency, and empower both consumers and local manufacturers. Ultimately, this will contribute to promoting environmental responsibility and establishing EcoBazaar as a leading platform for sustainable digital commerce.

Solution

To address the critical gap in accessing authentic, affordable, and verified eco-friendly products, we propose **EcoBazaar**, a sustainability-driven e-commerce platform that offers:

Eco-Friendly Product Discovery and Verification

Using a structured eco-verification workflow, the system evaluates products based on material composition, biodegradability, carbon footprint, ethical sourcing, and packaging standards. This ensures that only genuinely sustainable products are listed, eliminating greenwashing and helping users make trustworthy, environmentally responsible choices.

Intelligent Recommendation and Sustainability Insights

By analysing user preferences and sustainability metrics, EcoBazaar provides personalised eco-product recommendations. The platform also displays impact indicators such as estimated waste reduction and carbon savings, giving users meaningful insights into the environmental benefits of their purchases.

Transparent Product Information with Eco-Scores

Each product page features a detailed eco-score, life-cycle attributes, and material breakdown, enabling users to compare items based on sustainability factors rather than just price or brand. This transparency enhances decision-making and promotes conscious consumption.

User-Friendly Web Platform with Seller Support

Built using React.js/Next.js, the system includes smooth navigation, efficient product search, secure checkout, and multilingual accessibility. Simplified seller onboarding and inventory management tools empower smallscale artisans and local manufacturers to participate in the digital marketplace with minimal technical barriers.

Scalable, Secure, and Cloud-Ready Architecture

EcoBazaar's backend, developed using Node.js and MongoDB, supports high concurrency, real-time updates, and secure transactions. The platform is designed for cloud deployment, enabling consistent performance across urban and rural regions, with the ability to scale as user demand grows.

This solution ensures that EcoBazaar becomes a transparent, reliable, and sustainability-focused digital ecosystem—supporting consumers in making environmentally responsible choices and empowering genuine eco-friendly producers to reach a wider audience.

Key Components of the Proposed Solution

Product Data Standardization and Sustainability Profiling

All products submitted by sellers undergo structured data preprocessing, including material classification, biodegradability assessment, certification verification, and packaging evaluation. This ensures uniform sustainability profiling across the platform and enhances the reliability of eco-scores presented to users.

Eco-Verification Engine & Sustainability Scoring Algorithm

A rule-based and metric-driven evaluation system examines product attributes to generate a comprehensive sustainability score. Factors such as carbon footprint, recyclability, ethical sourcing, and production impact are computed to enable transparent comparison among eco-friendly products.

Explainability & Transparency Module

The platform provides detailed sustainability breakdowns, including origin tracking, material explanations, and environmental impact estimations. These transparent insights empower users to understand *why* a product is considered sustainable, fostering trust and reducing greenwashing.

Web-Based Eco-Commerce Platform

A secure, intuitive, and scalable interface—built using Next.js/React—allows users to browse product categories, view eco-scores, compare environmental impact, track their carbon savings, and complete purchases seamlessly. Sellers can upload products, manage inventory, and monitor sales through dedicated dashboards.

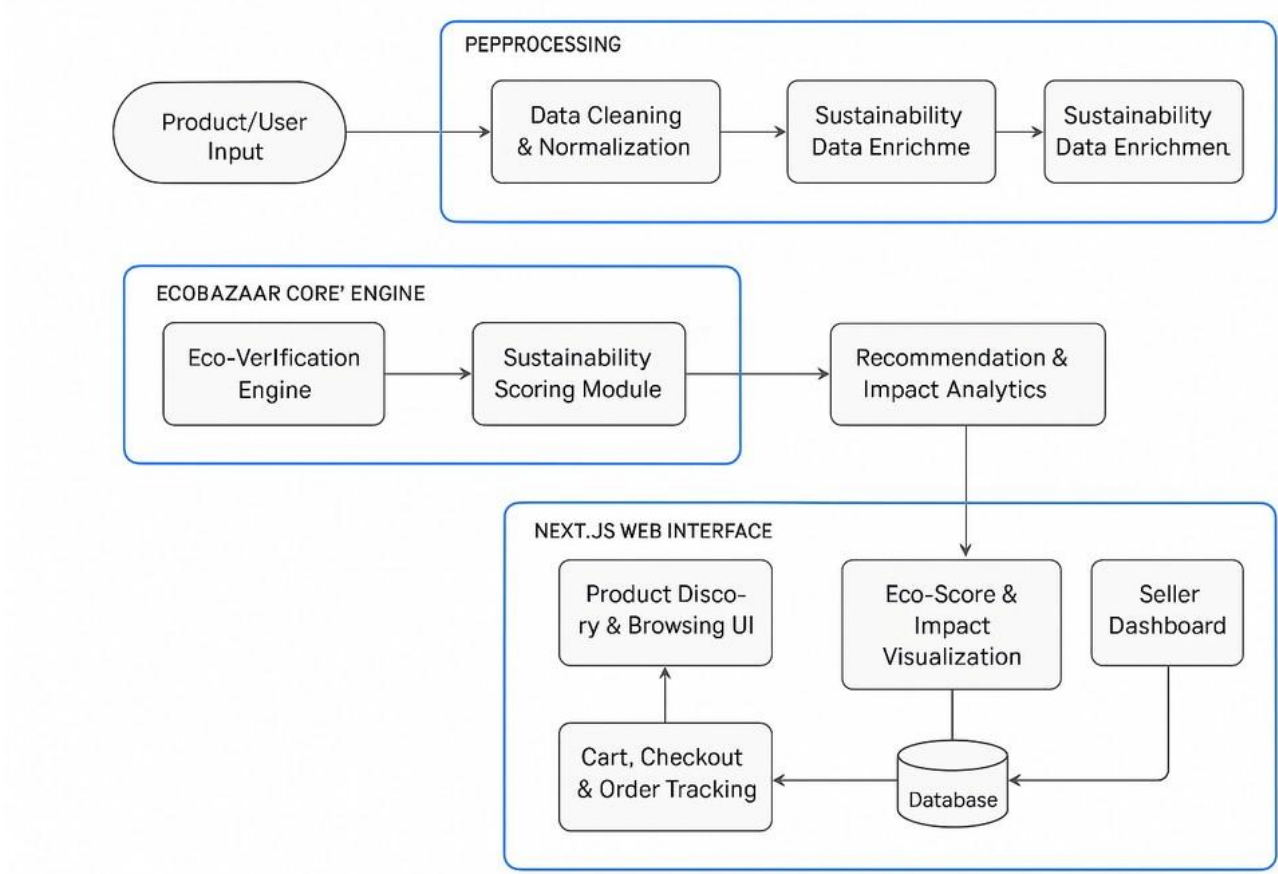
Continuous Data Improvement & Marketplace Expansion

The platform continuously integrates new sustainability research, updates verification rules, and incorporates user/seller feedback to refine eco-scoring accuracy. As the marketplace grows, new categories (e.g., zero-waste lifestyle products, sustainable fashion, natural wellness items) are added to expand eco-conscious choices.

Future Extensions

Planned advancements include AI-based green product recommenders, blockchain-enabled supply chain transparency, AR-based product visualization, multilingual support, and mobile application deployment to improve accessibility and create a more interactive and trustworthy sustainability ecosystem.

SYSTEM ARCHITECTURE: (Block Diagram)



Literature Review

1. Sustainable E-commerce Platforms

Model Name: *EarthHero – Sustainable Online Marketplace*

Published Year: 2017

Explanation: EarthHero is an online marketplace dedicated entirely to eco-friendly, zero-waste, and sustainably sourced products. It provides curated product listings verified for material safety, carbon impact, and ethical sourcing. Its model demonstrates the viability of specialised green e-commerce systems.

Advantages:

- Builds consumer trust through strict sustainability standards.
- Provides transparent product information and certifications.

Disadvantages:

- Primarily focused on US markets, limiting global accessibility.
- Higher product pricing may discourage budget-conscious consumers.

2. Green Consumer Behaviour Models

Model Name: *Theory of Planned Behaviour (TPB) in Green Purchasing*

Published Year: 1991 (widely adopted in sustainability research)

Explanation: TPB-based studies examine how attitudes, social influence, and perceived behavioural control shape consumer willingness to purchase eco-friendly products. These findings help platforms like EcoBazaar design features that encourage sustainable decision-making.

Advantages:

- Strong predictor of eco-conscious purchasing habits.
- Helps structure digital nudges for sustainability awareness.

Disadvantages:

- Does not fully address emotional influences.
- Consumer willingness may not always translate into action.

3. Eco-Labeling and Product Certification Systems

Model Name: *EU EcoLabel / USDA Organic Certification Frameworks*

Published Year: 1992 / 2000

Explanation: Global eco-certification systems define strict criteria for biodegradability, organic sourcing, and lifecycle sustainability. They influence digital marketplaces by providing verification standards for product authenticity, reducing greenwashing, and enhancing sustainability scoring.

Advantages:

- Provides formal, trusted evaluation of eco-product claims.
- Reduces misinformation and supports transparent marketplaces.

Disadvantages:

- Certification procedures can be expensive for small businesses.
- Different regions use varying standards, creating inconsistency.

4. Recommendation Systems in E-commerce

Model Name: *Content-Based and Collaborative Filtering Recommenders*

Published Year: 2000s (widely established)

Explanation: Modern e-commerce platforms use hybrid recommendation algorithms that analyse user behaviour, preferences, and item features to suggest relevant products. EcoBazaar applies these techniques to personalise sustainable product discovery based on eco-scores, material types, and user interests.

Advantages:

- Enhances user engagement and increases platform retention.
- Helps consumers discover eco-friendly alternatives effortlessly.

Disadvantages:

- Can suffer from over-filtering (“filter bubble” effect).
- Accuracy depends heavily on the quality of user and product data.

5. Blockchain for Supply Chain Transparency

Model Name: *IBM Food Trust / Provenance Blockchain Models*

Published Year: 2017

Explanation: Blockchain-based supply chain models enable transparent tracking of product origins, materials, and manufacturing processes. These approaches inspire possible EcoBazaar extensions that allow users to verify sustainability claims securely.

Advantages:

- Increases trust through immutable data records.
- Prevents tampering with eco-certification information.

Disadvantages:

- High implementation cost and computational overhead.
- Requires adoption across suppliers to function effectively.

6. Sustainable Marketplace Studies

Model Name: *Eco-Product Marketplace Adoption Studies*

Published Year: 2021–2023

Explanation: Recent research highlights the growing demand for platforms that consolidate ecofriendly products and provide clear sustainability metrics. Studies indicate that accessibility, transparency, and price fairness are key drivers for user adoption of green marketplaces like EcoBazaar.

Advantages:

- Validates market need and consumer interest in eco-commerce.
- Highlights design principles that improve platform effectiveness.

Disadvantages:

- Market competition is increasing in the sustainability domain. User
- trust heavily depends on consistent verification quality.

Feasibility Study

Technical Feasibility

Rapid advancements in modern web technologies and sustainable data processing frameworks make the development of EcoBazaar highly feasible. Robust frontend libraries such as **React.js** and **Next.js** support fast, responsive, and scalable user interfaces, while backend technologies like **Node.js** and **Express.js** allow secure seller onboarding, product verification, and order management workflows. Databases such as **MongoDB** and **MySQL** efficiently handle diverse product attributes, sustainability metrics, and user behaviour analytics. Additionally, integrating sustainability scoring and eco-verification is technically supported by rule-based engines, API-driven data enrichment systems, and third-party carbon footprint calculators. Cloud platforms such as **AWS**, **Render**, or **Firebase** provide reliable hosting, auto-scaling, and continuous deployment support. Overall, the technical components required to build, integrate, and maintain EcoBazaar are highly accessible, well-documented, and compatible with modern e-commerce architectures.

Operational Feasibility

EcoBazaar's operational demands are minimal compared to traditional retail systems. The platform can be deployed on standard cloud servers without requiring specialised hardware. Sellers can easily manage product uploads and inventories using intuitive dashboards, while users interact with a familiar and simple web interface requiring no technical expertise. The onboarding process for artisans and small-scale manufacturers is designed to be supportive and streamlined, reducing operational friction. Additionally, the eco-verification module ensures consistency in sustainability assessment, reducing manual overhead. Maintenance activities such as product verification updates, order tracking, and customer support can be smoothly handled by a small administrative team. With proper training and structured workflows, EcoBazaar can be efficiently operated even in resource-constrained environments.

Economic Feasibility

EcoBazaar leverages **open-source technologies**, significantly reducing development and operational costs. Since most components—React, Node.js, MongoDB, sustainability scoring libraries—are free to use, the platform avoids expensive proprietary software licensing. Cloud hosting can be scaled based on usage, allowing

flexible cost management as the user base grows. The marketplace model also supports monetization through low commission rates, premium seller features, and sustainability certification services. For artisans and small businesses, EcoBazaar provides an affordable alternative to mainstream platforms that typically charge high listing fees or commissions. By connecting local producers directly to conscious consumers, the platform can stimulate economic growth while keeping operational expenses low. This makes the project economically viable for both developers and stakeholders.

Environmental & Social Feasibility

EcoBazaar directly aligns with global sustainability goals by promoting eco-friendly products, reducing plastic consumption, and encouraging responsible consumer behaviour. The platform supports local artisans and cottage industries, strengthening community-based economies and reducing reliance on mass-produced, high-impact goods. Visual sustainability metrics and eco-scores empower consumers to make informed choices, enhancing environmental literacy. The model contributes to reduced environmental degradation by increasing the availability and visibility of biodegradable alternatives. Socially, EcoBazaar fosters inclusivity by providing digital access to small manufacturers who traditionally struggle with market exposure.

Challenges and Mitigation

Potential challenges include ensuring accurate sustainability claims, seller compliance, and user trust. These risks can be mitigated through strict eco-verification protocols, periodic audits, and transparent scoring criteria. Technical challenges such as managing increased platform traffic, preventing fraudulent listings, or ensuring continuous uptime can be addressed through cloud scaling, advanced security practices, and database optimization. Adoption challenges from sellers may arise due to unfamiliarity with digital tools, mitigated through training resources, multilingual support, and simplified dashboards. Data privacy regulations can be met through encryption, compliance with data protection laws, and secure hosting practices.

Methodology/Planning of Work

Database Setup: Curate a comprehensive collection of chest X-ray images and associated metadata, including diagnoses, patient demographics, disease severity scores, and progression records. This database will form the backbone for training, validation, and similarity retrieval. Continuous updates from open-source medical repositories, such as CheXpert and ChestX-ray8, will ensure evolving accuracy and coverage for a wide range of pulmonary diseases.

Preprocessing Module: Implement advanced preprocessing workflows using OpenCV and allied libraries to optimize X-ray images prior to model analysis. Tasks will include resizing for uniformity, removing noise artifacts, and performing normalization across image inputs. The preprocessing module ensures consistent input quality and prepares the data for effective feature extraction by the AI models.

Similarity Engine: Develop a machine learning-driven engine using EfficientNet for extracting deep features from images, and integrate similarity algorithms to retrieve the most comparable cases from the curated database. This engine will support severity scoring and disease progression mapping, so clinicians can access relevant past cases and track patient changes over time. Advanced algorithms will be evaluated to ensure robust and reliable similarity detection for multiple lung pathologies.

Explainability Module: Incorporate Grad-CAM or similar methods to visualize model predictions and highlight affected lung regions directly on the X-ray images. Clinicians will benefit from transparent, interpretable outputs that indicate why specific cases are retrieved or certain diagnoses made. This module is critical for building trust and supporting informed clinical decisions.

Web Interface: Utilize Next.js to engineer a responsive, secure, and user-friendly platform for uploading chest X-rays, displaying retrieved similar cases, providing severity scores, and tracking disease progression. Accessibility features, multi-language support, and intuitive navigation will be prioritized to ensure usability across diverse healthcare settings, from urban hospitals to remote clinics.

Testing and Validation: Conduct thorough validation using k-fold cross-validation, expert radiologist feedback, and real-world usability scenarios. Testing will assess diagnostic accuracy, similarity retrieval effectiveness, interface usability, and system robustness against varied image qualities and disease types. Continuous feedback loops will be established to refine the system post-deployment and maintain clinical reliability.

Facilities Required

Software:

1. Operating System

- Windows / Linux / macOS (any of these can be used for development)
- Must support Node.js, MongoDB, Git, and development IDEs.

2. Backend Development

- **Node.js**
 - JavaScript runtime environment for building the backend services.
 - Handles APIs for user authentication, product management, cart, orders, and seller dashboards.
- **Express.js**
 - Web framework for Node.js to create RESTful APIs and route management for EcoBazaar.

3. Database

- **MongoDB (Local or Cloud – e.g., MongoDB Atlas)**

- NoSQL database to store:
 - User details
 - Seller profiles
 - Product catalog and attributes
 - Orders and transaction history
 - Sustainability scores and analytics data

4. Frontend Development

- **React.js**

- For building a dynamic, component-based, and responsive user interface of EcoBazaar.

- **Tailwind CSS / CSS3**

- For designing a clean, minimal, and mobile-friendly UI aligned with the eco-theme.

- **HTML5**

- Basic structure of all web pages.

5. AI & Model Integration (for Product Verification System)

- **Python (3.x)**

- For building and running the core AI model for product verification.

- **TensorFlow / PyTorch**

- To implement and fine-tune the **EfficientNet** model.

- **Grad-CAM Libraries**

- For generating visual explanations that highlight regions of product images contributing to “ecofriendly” classification.

- These models can be exposed through a **REST API service** and called from the Node.js backend.

6. Design & Prototyping

- **Figma / Adobe XD**

- For UI/UX design of:
 - Landing page
 - Product listing & details
 - Cart & checkout
 - Seller dashboard
 - Admin dashboard

7. Testing & API Tools

- **Postman / Thunder Client**

- For testing API endpoints (login, product CRUD, orders, etc.)

- **Jest / Mocha (Optional)**

- For unit and integration testing of backend logic.

8. Version Control & Collaboration

- **Git**

- For source code management.

- **GitHub / GitLab**

- For remote repository hosting, collaboration between team members, and maintaining project history.

9. Deployment & Hosting

- **Cloud Platform (e.g., Render / Vercel / AWS / Railway)**
 - For hosting backend (Node.js) and frontend (React) services.
- **Domain & SSL (Optional for Production)**
 - To provide secure (HTTPS) access to users.

10. Payment & Notification Integrations (If Implemented)

- **Razorpay / Stripe API**
 - For secure online payment processing for user orders.
- **Email / SMS Service (SendGrid, Twilio, etc.) – Optional**
 - For sending order updates, alerts, and notifications.

Expected Outcomes

1. Verified Authentic Eco-Friendly Products

With AI-backed verification using EfficientNet + Grad-CAM, EcoBazaar will ensure that only genuinely ecofriendly products are listed. This will reduce misleading claims (greenwashing) and build strong trust among consumers.

2. Empowerment of Local Artisans and Cottage Industries

EcoBazaar will provide digital visibility, fair marketplace access, and analytical tools to small-scale sellers and artisans, helping them increase income, reach new customers, and operate efficiently without requiring advanced technical skills.

3. Enhanced Consumer Trust Through Transparency

Customers will be able to see detailed sustainability information, including:

- Sustainability scores
- Verified eco-labels
- Material composition
- Eco-impact metrics such as carbon savings
- This transparency will encourage more responsible and informed buying decisions.

4. AI-Assisted Product Verification and Quality Control

With AI-based image classification and Grad-CAM visual explanations, the system will detect non-eco materials, plastic usage, or misleading product photos. This will significantly improve the accuracy and quality of the product catalog.

5. Sustainable Shopping Made Easy and Personalized

EcoBazaar's recommendation engine will suggest products based on user preferences, past purchases, and sustainability goals, making eco-friendly shopping more accessible and consistent for consumers.

6. Strengthened Environmental Impact Awareness

Features such as the GreenTrack carbon footprint meter will help users understand the environmental benefits of their choices, encouraging long-term adoption of sustainable habits.

7. Efficient Digital Marketplace Operations

The platform will streamline seller onboarding, product management, order processing, inventory tracking, and payment settlements. This will reduce operational friction and improve marketplace efficiency for sellers and administrators.

8. Improved Accessibility Across Urban and Rural Regions

With a lightweight, responsive design and scalable architecture, EcoBazaar will be accessible even in areas with limited digital infrastructure, promoting inclusive participation from diverse communities.

9. Increased Adoption of Eco-Friendly Lifestyle Products

As trusted, affordable, and verified sustainable products become easier to access, consumers are more likely to transition away from harmful or plastic-based alternatives, accelerating environmentally responsible consumption.

10. Contribution to Research, Innovation, and Green Commerce

The platform and its AI verification system can support future research in sustainability, eco-product classification models, and innovations in green retail, carbon scoring, and lifecycle assessment, positioning EcoBazaar as a foundation for technological and environmental progress.

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